

Automated Construction of Multi-Element MEAM Interatomic Potentials

A DFT-Informed Neural-Network Optimization Framework — Final Report

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PHYS 400 — Final Presentation





Two elastic constants govern the structural response of every load-bearing component:

- $E = \sigma/\varepsilon$
Young's modulus — uniaxial stiffness (GPa).
- $\nu = -\varepsilon_{\text{lat}}/\varepsilon_{\text{axial}}$
Poisson's ratio — transverse contraction.

For cubic crystals these derive analytically from the two independent stiffness constants C_{11} and C_{12} — the quantities targeted from first principles in this work.

Representative structural applications

Application	Material	E (GPa)	ν
Aerospace fuselage	Al 7075	71–72	0.33
Turbine disk	Inconel 718	200	0.29
Orthopaedic im-plant	Ti 6Al–4V	110	0.34
Automotive struc-ture	Mg AZ91	45	0.35
Structural steel	Mild steel	200	0.30
MEMS resonator	Si (single xtal)	130	0.28
Rolling-element bearing	52100 steel	210	0.30



Quantities governed by E :

- Beam deflection $\delta = FL^3/(3EI)$
- Euler buckling $P_{cr} = \pi^2 EI/(KL)^2$
- Natural frequency $\omega_n \propto \sqrt{E/\rho}$
- Hertzian contact pressure $\propto E^*$

Quantities governed by ν :

- Bulk modulus $K = E/[3(1 - 2\nu)]$
- Plane-stress vs. plane-strain regimes
- Stress concentration at geometric discontinuities
- Fracture toughness $\propto (1 - \nu^2)$

Representative ranges

Material	E (GPa)	ν
Polymers	0.5 – 3	0.40
Mg alloys	40 – 45	0.35
Al 7075	70 – 75	0.33
Ti 6Al-4V	110	0.34
Steel	200	0.30
Rubber	~ 0.01	$\rightarrow 0.5$

Mechanical stability criteria

Isotropic continuum: $-1 < \nu < 0.5$.

Cubic crystal (Born–Cauchy): $C_{11} > C_{12}$.



Interatomic potential. Closed-form approximation of the many-body potential energy $U(\{\mathbf{r}_i\})$; forces obtained as $\mathbf{F}_i = -\nabla_i U$.

Historical development:

- Pairwise (Lennard–Jones, Morse) — inadequate for metallic bonding.
- Embedded Atom Method (Daw & Baskes, 1984) — suitable for FCC metals.
- **Modified EAM** [1, 2]: introduces angular partial densities $\rho_i^{(l)}$, $l = 1-3$; established formalism for BCC, HCP and mixed-bonding metals.

Density-functional theory (DFT).

First-principles electronic-structure method, formulated via the Kohn–Sham equations [3] and implemented here with a plane-wave + pseudopotential basis (QE, Giannozzi et al. 4).

Quantities computed:

- Cohesive energy E_{coh}
- Equilibrium lattice a_0 , bulk modulus B
- Binary formation energies $E_{\text{form}}(i, j)$
- Single-crystal elastic constants C_{11} , C_{12}

DFT: high accuracy, limited scale ($\leq 10^3$ atoms).
MEAM/MD: empirical, large scale ($\geq 10^6$ atoms).

DFT provides the reference dataset against which the MEAM potential is calibrated.

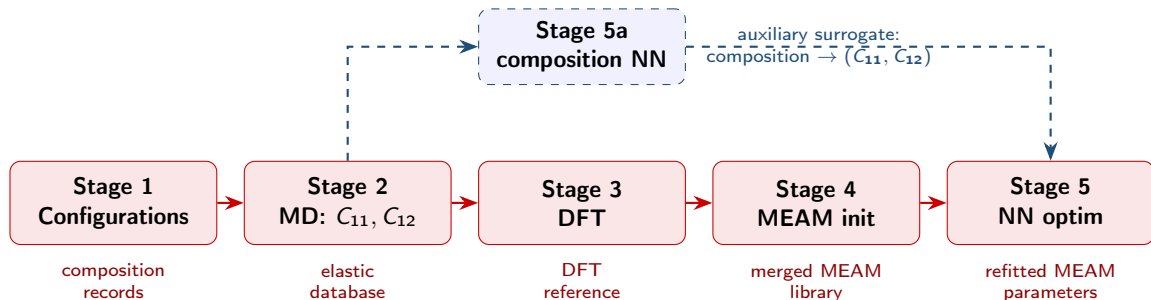


Published MEAM libraries cover disjoint subsets of elements; no single file spans the nine elements specified by Al 7075.

Library	Al	Co	Cr	Cu	Fe	Mg	Mn	Mo	Ni	Si	Ti	Zn
AlMgZn	✓					✓						✓
AlSiCuMgFe	✓			✓	✓	✓				✓		
CuMo				✓				✓				
FeMnNiTiCuCrCoAl	✓	✓	✓	✓	✓		✓		✓		✓	
Merged (this work)	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓

Limitation of direct merging: cross-interaction parameters between elements drawn from distinct libraries are not defined; manual refitting does not scale to $\binom{12}{2} = 66$ binary pairs.

Pure machine-learning potentials [5–7]: require $10^4 - 10^5$ DFT configurations and forfeit the physical interpretability of MEAM.



The four published libraries are merged into a single 12-element file; cross-terms are refitted by inverting an NN surrogate of the MEAM $\rightarrow (C_{11}, C_{12})$ map under L-BFGS, with single-element parameters anchored to DFT references.



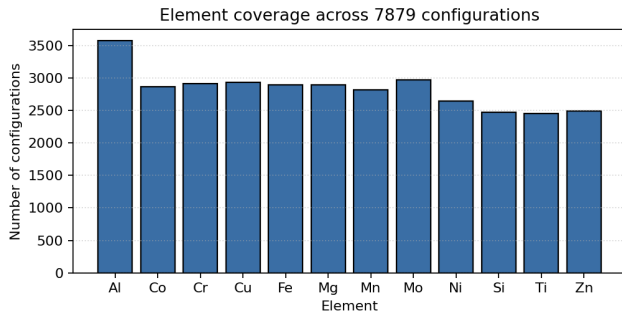
Sampling.

Uniform draws on the 12-element composition simplex (Al, Co, Cr, Cu, Fe, Mg, Mn, Mo, Ni, Si, Ti, Zn);
Al-weighted to oversample the aluminium-alloy region.

Corpus statistics.

~7900 configurations generated;

7815 retained after the physicality filter ($E > 0$, $0 < \nu < 0.48$, $C_{11} > C_{12}$).



Element-occurrence distribution over the retained corpus.



Implementation. [8]

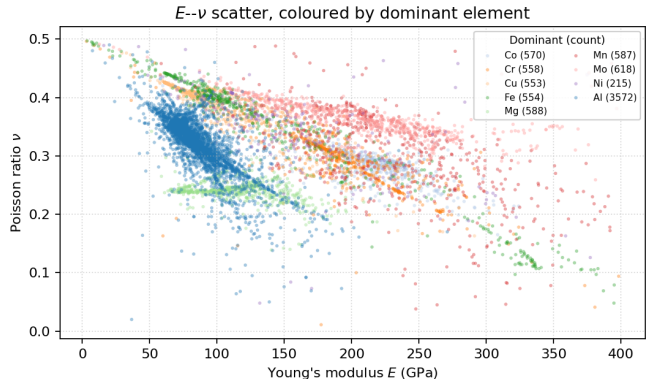
Custom LAMMPS build (MEAM species limit raised 5 $\rightarrow$ 20).

Per-configuration protocol:

- ~ 4 nm cubic supercell
- Atom-type reassignment to composition $\{c_i\}$
- Anisotropic energy minimisation
- Axial strain $\delta = 10^{-3}$ applied along $\pm x, \pm y, \pm z$
- Central-difference $\rightarrow C_{11}, C_{12}$

	Median	Std.	Range
E (GPa)	103.6	69.1	3 – 398
ν	0.326	0.063	0.01 – 0.50

$N = 7815$ retained configurations.



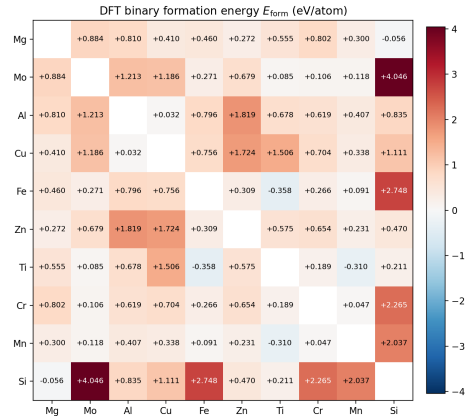
E - ν distribution, coloured by dominant element. Mg- and Zn-rich compositions cluster near $\nu \rightarrow 0.5$, motivating training on C_{11}, C_{12} rather than (E, ν) .

Stage 3: Quantum Espresso DFT References



Element	Lat.	a_0 (Å)	E_{coh}	B (GPa)
Al	FCC	4.047	3.611	77.5
Cu	FCC	3.639	3.857	138.5
Si	DIAMOND	5.471	5.130	89.3
Ti	HCP	2.915	6.284	114.5
Zn	HCP	2.766	0.964	68.4
Cr	BCC	2.848	3.964	213.1
Fe	BCC	2.806	6.834	420.5
Mg	HCP	3.177	1.483	47.8
Mn	BCC	2.803	3.728	225.7
Mo	BCC	3.169	10.353	280.8

Elemental references (E_{coh} in eV/atom; Birch–Murnaghan fits).



Binary formation energies $E_{\text{form}}(i, j)$ (eV/atom). Negative values indicate exothermic mixing; positive values indicate limited mutual solubility. These references seed the MEAM cross-term parameters.



Screening exponent from the DFT triple (a_0 , E_{coh} , B):

$$\alpha = a_0 \sqrt{\frac{9 B \Omega}{E_{\text{coh}}}}$$

Ω : primitive-cell volume.

Together with the DFT-seeded cross-term energies $E_c(i, j)$, these constants yield a complete 12-element MEAM library suitable for direct evaluation in LAMMPS.

Computational cost: ~ 70 ms once DFT references are cached. The α -Mn equation of state did not converge; a relaxed BCC approximation is used.

Element	Lat.	a_0 (Å)	E_{coh}	Ω (Å ³)	α
Al	FCC	4.047	3.611	16.57	4.470
Cu	FCC	3.639	3.857	12.05	4.930
Si	DIAMOND	5.471	5.130	20.46	4.473
Ti	HCP	2.915	6.284	35.02	5.987
Zn	HCP	2.766	0.964	29.93	10.921
Cr	BCC	2.848	3.964	11.55	5.906
Fe	BCC	2.806	6.834	11.05	6.179
Mg	HCP	3.177	1.483	45.36	9.063
Mn	BCC	2.803	3.728	11.01	6.120
Mo	BCC	3.169	10.353	15.91	4.924

Stage 5a: Composition $\rightarrow (E, \nu)$ Neural Surrogate



Architecture. $12 \rightarrow 32 \rightarrow 20 \rightarrow 2$
fully-connected MLP
with ReLU activation; Huber loss on
 $\hat{C}_{11}$, $\hat{C}_{12}$;
deep ensemble of five seeds [9] for
epistemic uncertainty.

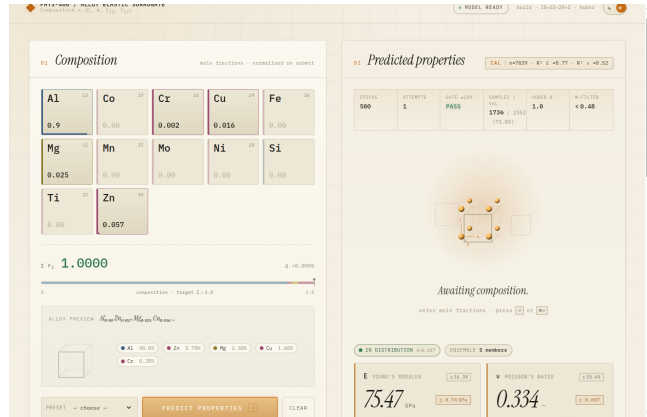
Held-out performance (30 % split):

- $R_E^2 = 0.77$
- $R_\nu^2 = 0.52$

Validation on the Al 7075 preset:

$\hat{E} = 75.5 \text{ GPa}$, $\hat{\nu} = 0.334$
(published reference: 71–72 GPa, $\nu \approx 0.33$).

Distributed as a portable application image.



Interactive composition predictor with k -nearest-neighbour distance-based out-of-distribution flagging.



Closed-loop procedure:

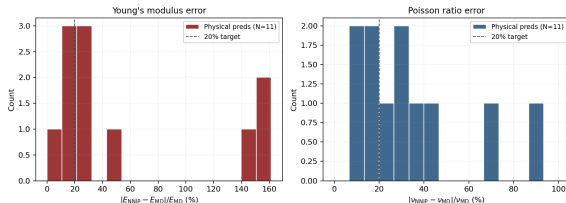
- 1 NN surrogate learns (MEAM params) $\rightarrow (C_{11}, C_{12})$.
- 2 L-BFGS [10] minimises the Huber loss through the surrogate.
- 3 Optimised parameters are re-evaluated in LAMMPS; new samples are appended and the surrogate retrained.

Atomic checkpointing renders the procedure resumable.



Surrogate Huber-loss decay from 1.0 to 0.16 over 10^4 steps.

NNIP Stage-5 verification on 30 held-out alloys | E within 20%: 4/30 | ν within 20%: 4/30 | unphysical NNIP outputs: 19/30



Held-out alloys / pass stability filter	30 / 11
Within 20 % of MD on E / on ν	4 / 4
Median error on E / ν (physical)	23 % / 26 %

Verification outcome.

Of the 30 held-out alloys, 19 violate the Born–Cauchy stability criterion. Among the 11 stable predictions, 4 satisfy the 20 % engineering tolerance on both E and ν .



Contributions

- MD elastic database, 7815 alloys
- DFT reference set, 10 elements
- Merged 12-element MEAM library
- Composition surrogate ($R_E^2 = 0.77$, deployed)
- Resumable Stage 5b framework

Current limitations

- 4/30 predictions within tolerance
- 19/30 violate Born–Cauchy
- Cross-term parameters saturate bounds

Identified causes

- Under-determined inverse problem (~ 200 parameters, ≤ 100 targets)
- Surrogate–LAMMPS residual mismatch
- Stability constraints absent from the loss

Planned extensions

- Restrict optimisation to single-element-anchored cross-terms ranked by Sobol sensitivity
- Active-learning expansion to ≥ 500 medoids
- Physics-informed loss with stability penalties
- SQS DFT references [11] at intermediate compositions



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